

Social Media Content Generation: An AI-Based System for Automated Viral Short-Form Video Creation

Gautham B. and K Somasundaram

Department of Mathematics

Amrita School of Physical Sciences, Coimbatore

Amrita Vishwa Vidyapeetham, India

gauthambalraj07@gmail.com; s_sundaram@cb.amrita.edu

Abstract—The prevalence of short-form video content on platforms such as YouTube Shorts and Instagram Reels has prompted the need for automated tools for extracting and optimizing viral moments from long-form video. This work outlines a fully realized LLM-based system that automatically identifies, processes, and augments viral video clips. The system combines yt_dlp for high-definition video extraction, AssemblyAI for speech-to-text transcription with accurate word-level timestamping, and LLM based workflow to identify high-engagement points. MoviePy and OpenCV are also used for resizing, formatting, and overlaying synchronized, word-level highlighted subtitles to maximize viewer engagement. By removing the need for manual editing, this system allows content creators and digital marketers to effectively repurpose long-form videos into high-impact, platform-optimized clips. This paper delves into the architecture of the system, challenges of implementation, and its future uses in the changing context of AI-based content generation and social media advertising.

Index Terms—AI-Powered Content Generation, Social Media Automation, Short-Form Video Processing, Natural Language Processing (NLP), Video Summarization, Deep Learning for Video Editing, Automated Subtitle Generation, LLM for Content Analysis, Computer Vision in Video Processing, AI-Driven Digital Marketing, Content Repurposing, Viral Video Extraction.

I. INTRODUCTION

The dramatic rise of short-form video platforms like YouTube Shorts, Instagram Reels, and TikTok has transformed digital content viewing. These sites have triggered a shift by focusing on short, emotionally charged content that is easily consumed and readily shareable. Manually finding and editing fun parts of long-form content, however, is still a time-consuming and labor-intensive process. Scalable automation has never been more essential for marketers, media organizations, and content creators.

To solve this problem, we introduce an AI-powered platform for auto-generating short-form videos from lengthy YouTube videos and making them go viral. Our platform is capable of automating the end-to-end process from downloading the video and providing a transcript of the audio, to determining high-engagement points and editing the video with synchronized subtitles, optimized for platforms such as TikTok and YouTube Shorts.

Our system architecture incorporates several technologies. yt-dlp is utilized to pull media content, and AssemblyAI speech-to-text transcribes dialogue with word-level timestamping, allowing correct alignment between video frames and dialogue. A Multimodal Large Language Model (LLM) detects "viral" moments as a function of semantic density, emotional tone, and relatability to the viewer, akin to methods investigated in LAVCap, where LLMs integrate audio-visual signals for semantic captioning [14]. Motivated by VG-TVP [15], our method conducts visually grounded prompting that aligns visual and text modalities to maintain procedural coherence across video segments.

This architecture is particularly sensitive to signals of engagement like humor, surprise, and relatability central to the drivers of virality as explored in the generative survey by Zhou et al. [1]. It diverges from classic heuristic editing systems in that it uses zero-shot reasoning to infer the content at a narrative and emotional level, similar to the personalized LLM ranking methods in LLMRS [13].

After clip selection, the system uses MoviePy and OpenCV to edit the video, such as converting to 9:16 aspect ratio and superimposing word-synced subtitles a method found to enhance viewer engagement in mobile-first platforms. Lastly, we use a web-based Flask interface where users can enter any YouTube URL and obtain AI-generated short clips to share.

Our design is also inspired by knowledge-based systems like TravelRAG, in which domain knowledge graphs are employed to enhance content accuracy and minimize hallucination in LLM-generated text [17]. Though our execution is specific to video content, retrieval-augmented generation (RAG) principles are helpful in providing contextual filtering as well as generating metadata.

Major Contributions of This Work:

- Construction of a full-stack end-to-end automated system for short-form video production using LLM-based reasoning.
- Multimodal transcript analysis integration with word-level timestamping for clip segmentation accuracy.
- Utilization of audio-visual fusion and semantic alignment towards viral moment detection [14], [15].

- Mobile-first optimized dynamic subtitle generation enhancing engagement and accessibility.
- Extensible and scalable system utilizing LLM insights and user-centered outputs in line with ongoing trends in retrieval-augmented and knowledge-enhanced generation [13], [17].

Through the automation of one of the most labor-intensive stages of content creation, our system provides a scalable, intelligent, and easy-to-use solution to content producers, teachers, marketers, and agencies. The suggested architecture unifies the most recent breakthroughs in LLMs, multimodal learning, and retrieval-based generation into an integrated pipeline that optimizes content impact while reducing human effort.

II. LITERATURE REVIEW

With the introduction of short video platforms such as YouTube Shorts, Instagram Reels, and TikTok, content generation using AI has become a major area of interest as a research topic. Various approaches have been attempted to enhance video understanding, summarization, and content generation using Large Language Models (LLMs) and deep learning techniques.

Traditional video temporal localization methods have been focused on event boundary timestamp prediction. Subsequent approaches have leveraged the application of LLMs further to accurately record semantic comprehension. MeCo, introduced in Measure Twice, Cut Once [7], replaces event detection from timestamps with end-to-end event segmentation so that LLMs are able to best utilize their semantic reasoning abilities. By integrating the application of a query-focused captioning task, MeCo bridges the gap between high-level video understanding and event localization. This semantic-centered approach has been proven to work better than boundary-centered approaches. Similarly, VideoLLM [6] proposes a video sequence modeling model with LLMs, transforming visual data into tokenized representations for end-to-end video reasoning. The model aligns video understanding with the sequence modeling capacity of LLMs, enabling improved generalization across diverse video understanding tasks.

Generative AI has significantly enhanced video analysis by automating content creation, summarization, and user interaction. A Survey of Generative AI and LLM for Video Generation, Understanding, and Streaming [1] presents a wide-ranging overview of how LLMs are enhancing video understanding through recognizing key moments and creating summaries. The models facilitate the engagement of users and stream videos better through adapting the transmission of contents based on the interest of the users. Furthermore, self-managed video summarization has emerged into attention as well, particularly in specific application areas such as cooking videos [4], sports [12], and healthcare communication [9]. Transformer models along with convolutional neural networks (CNNs) have been utilized for creating short summaries of recipes for cooking videos. Implicit Embedding-Based Multi-Modal Attention Network [12] similarly utilizes user prefer-

ence and attention-based on audio for dynamically summarizing cricket sports videos.

Video content creation has also been automated in the domain of personalized and interactive media. Hashtag generation for short videos [10] demonstrates the ability of AI-based tools to enhance content discoverability and interaction. With multimodal signals, AI can generate dynamic hashtags, increasing video reach and visibility on social media platforms. In addition, AI-Based Chinese-Style Music Generation from Video Content [3] explains how cross-modal content creation can utilize generative models to enable AI to generate culturally contextualized music tailored to suit video looks. Such developments highlight the versatility of AI in generating various forms of multimedia content, from video highlights to personalized soundtracks.

Apart from entertainment and content creation, AI video generation has its application in medicine. Overcoming Elderly's Medical Communication Problem with Personalized LLM-Generated Short-Form Videos [9] outlines an approach where LLMs generate instructional videos for improving the medical communication of elderly patients. Integrating persona modeling and NLP, these videos bypass cognitive impairments and improve medical instruction comprehension.

Apart from the above-mentioned research, new developments in multimodal AI systems also highlight the revolutionary impact of Large Language Models (LLMs) in content generation pipelines. One such major contribution is Aditi Singh's in-depth survey on AI Text-to-Image and Text-to-Video Generators [16]. This publication describes how generative models like DALL-E, CogView2, and Imagen utilize deep learning and NLP methods to convert textual inputs into high-quality images and videos. The paper compares and categorizes these models in terms of architectural design, training corpus, generation quality, and interpretability. It further considers real-world applications in marketing, education, and entertainment highlighting their applicability to generating social media content. Although these models primarily revolve around media generation from text input, they closely relate to the core objectives of systems like ours that intend to transform long videos into short but captivating clips based on insightful text.

Another significant piece of work is LAVCap: LLM-based Audio-Visual Captioning using Optimal Transport by Rho et al. [14], which presents an audio-visual captioning model that combines audio and visual inputs for better semantic alignment and text output. The LAVCap model resolves the issue of multimodal feature combination by adding an Optimal Transport Attention Module, which better aligns audio and visual feature spaces compared to traditional cross-attention methods. Specifically, the system beats previous state-of-the-art approaches on the AudioCaps dataset without large-scale pre-training or post-processing. For short-form video production, particularly for noisy or complicated scenes, such cross-modal fusion methods can be of great use in producing context-rich captions that enhance viewer comprehension and engagement.

Expanding LLM applicability into recommendation systems, LLMRS: Unlocking Potentials of LLM-Based Recommender Systems for Software Purchase by Angela John et al. [13] presents a zero-shot recommender system that employs pretrained LLMs to decode user reviews and provide personalized recommendations. Their approach applies MPNet embeddings to read user inputs and determine cosine similarity with software product descriptions. This model outperforms standard collaborative and content-based filtering approaches without extensive training, evidencing the generalization capability and strength of LLMs. This research proves the potential of LLMs in comprehending unstructured, user-created content a central difficulty in sentiment evaluation of viewers and adjusting social video output to audience interests.

In the travel industry, TravelRAG by Song et al. [17] introduces a knowledge graph-based information retrieval paradigm that integrates Retrieval-Augmented Generation (RAG) with a multi-layered knowledge graph. TravelRAG facilitates precise and current tourist recommendation generation through dynamically structuring social media data into knowledge representations. It minimizes hallucination and enhances contextual precision in LLM responses. Since our system also concerns managing and summarizing complicated audiovisual material into readable brief videos, the RAG method with a formalized memory (such as video summary metadata) presents a promising avenue for enhancing factual grounding in content produced by LLM.

Finally, VG-TVP: Multimodal Procedural Planning via Visually Grounded Text-Video Prompting by Ilaslan et al. [15] presents a system that produces multimodal procedure plans textual descriptions coupled with synthetic videos for goal-directed activities such as cooking. The model applies a combination of video-to-text and text-to-video bridges and a Fusion of Captioning (FoC) mechanism to generate temporally aligned, human-oriented procedural guides. VG-TVP emphasizes the possibility of LLMs to manage sophisticated, multimodal content creation pipelines. Its capacity to create coherent visual plans consistent with user commands or objectives is particularly appealing to the aspirations of systems translating instructional or narrative-length videos into viral, short-form versions that meet viewers' requirements.

Collectively, these projects showcase the rich and fast-changing landscape of LLM use cases. From captioning to recommendation, procedural planning to domain-specific retrieval, these systems provide tangible architectural knowledge and performance standards that shape and augment AI-driven short-form video content generation system design.

III. METHODOLOGY

This work presents an AI-based framework for end-to-end automated short-form video generation, capable of intelligently identifying viral moments from long-form YouTube videos. The system proposed in this work combines speech recognition, vision-language modeling, semantic fusion, and large language models (LLMs) to produce shareable clips

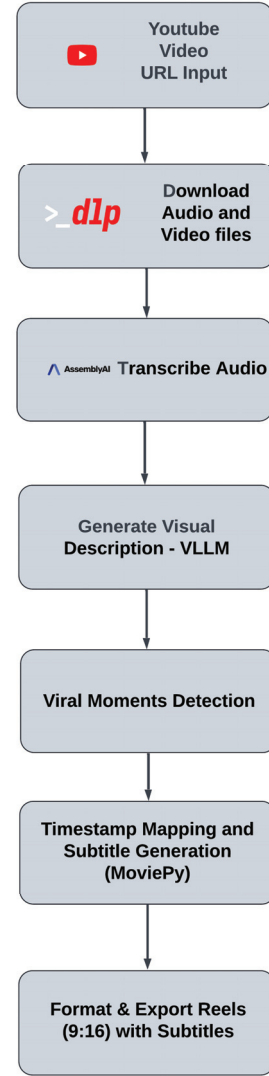


Fig. 1. Workflow

optimized for today's social media. Every element of the workflow illustrated in Figure 1 has been designed to maximize scalability, contextual depth, and engagement.

A. Video and Audio Retrieval

The first step of the workflow is to download video and audio content independently in order to preserve the maximum possible fidelity for downstream processing. This is done with the help of the `yt_dlp` library, a new and feature-rich fork of the widely used `youtube-dl` utility. `yt_dlp` brings important improvements over its ancestor, such as improved support for adaptive bitrate streaming formats (like DASH and HLS), better metadata extraction functionality, post-processing hooks that can be customized, and faster multi-threaded downloads.

The download process is controlled by function that takes parameters like URL of the long format video, target resolution, desired encoding format, and download concurrency bounds. In order to achieve strength, the function comes with

automatic retry measures, graceful network interruption handling, and smart fallback policies when faced with unsupported formats.

By intentionally downloading the video and audio streams independently instead of using merged formats, the system maintains the original quality of every modality. The separation is important for downstream operations, where high-fidelity audio is required for accurate Automatic Speech Recognition (ASR) through AssemblyAI, and uncompressed video frames are required for efficient analysis during the visual description creation process.

In addition, the modularity of this retrieval module guarantees compatibility with a variety of online video sites other than YouTube, providing flexibility for future system extensions.

B. Audio Transcription Using AssemblyAI

Precise and timely speech-to-text translation is an essential requirement for strong content analysis and viral moment identification. This is achieved by the system using AssemblyAI, a state-of-the-art Automatic Speech Recognition (ASR) platform renowned for its high-quality transcription, sophisticated linguistic modeling, and scalability for real-time deployment.

AssemblyAI offers enriched transcription outputs, such as word-level timestamps, filler word removal (e.g., "uh", "um") automatically, punctuation restoration, speaker diarization, and paragraph segmentation. Such advanced features considerably improve both readability and structural alignment of the transcripts, making downstream tasks like subtitle generation, scene segmentation, and semantic content extraction easier.

The transcription pipeline is composed of two tightly coupled functions. First, transcribe submits the extracted audio stream to AssemblyAI's asynchronous API and returns a structured transcript object. This object encapsulates all recognized text and associated metadata (e.g. confidence scores), making it immediately consumable by the next stage. The `extract_word_level` function then ingests that transcript object and deconstructs it into a precise sequence of word-level entries, each annotated with accurate start-and-end timestamps.

This high-resolution temporal alignment guarantees that each uttered word can be traced back directly to its respective location in the video timeline. This level of detailed timestamping is crucial not only for subtitle synchronization but also for precisely matching identified viral content to particular video frames during clip extraction.

By using the high-fidelity transcription feature of AssemblyAI, the system is able to ensure that the language foundation of the video is acquired with optimal semantic detail and timing accuracy and thus offers a sound basis for following multimodal fusion and viral moment detection.

Moreover, since the transcription step operates independently within the workflow, AssemblyAI can be easily replaced with alternative speech-to-text models such as OpenAI's Whisper, Coqui TTS, XTTS-v2, Tortoise TTS, or other open-source solutions. As long as the output adheres to the expected transcript structure, it can be seamlessly passed to

the `extract_word_level` function without requiring any modifications. This modular approach ensures that downstream processing remains consistent regardless of the transcription engine used.

C. Video Segmentation and Description Generation

After the audio transcription is complete, the system proceeds to structurally segment the video into a series of shorter clips. Each of these segments is then processed through a multimodal large language model (LLM) to extract rich scene-level semantic information through visual analysis.

This Process is employed to generate detailed descriptions of each segment, capturing a diverse range of visual cues, including:

- **Action Recognition:** Recognition of important human actions and interactions within the frame.
- **Object Detection:** Enumerating important objects and props that might contribute to thematic context.
- **Emotion Recognition:** Determining the emotional tone from facial expressions, body language, and scene composition.
- **Scene Change Detection:** Detecting sudden cuts, visual surprises, or environment changes that signal possible peak engagement points.

These per-segment visual abstracts complement the captured audio modality in transcription, developing a more comprehensive comprehension of every clip. Cues non-verbal like an abrupt change of setting, a comedic gesture, or an exaggerated visual response tend to be most influential in segment potential virality even if these types of elements are never mentioned verbally in the dialogue.

By creating multimodal semantic representations at the segment level, the system markedly improves the capabilities of the system to detect and interpret micro-level engagement signals that would otherwise be missed based on transcript-level analysis. This richer representation later is used for multimodal detection of viral moments to ensure that both verbal and visual aspects are utilized to the same extent in detecting high-impact short-form content.

D. Multimodal Fusion for Segment Enrichment

To achieve a more comprehensive understanding of each short-form segment, the system performs multimodal fusion by integrating information from two distinct but complementary sources:

- The corresponding audio transcription derived from AssemblyAI's Automatic Speech Recognition (ASR) system.
- The visual scene description generated by multimodal LLM for each segment.

This integration process yields an enriched semantic representation for each segment, both capturing the overt verbal message and the covert visual storytelling aspects.

In embracing the visual modality, the system guarantees that essential non-verbal events including silent motion, emotional face, visual joke, scene cuts, and active motion are properly

identified and incorporated into identifying viral moments. These aspects mostly determine audience responses but are hard to discern through the audio transcript alone.

The fusion is done by synchronizing the temporally aligned transcript and the visual description into a combined structured input. This enriched multimodal segment is the foundation for the follow-up viral moment detection phase so that the system can reason simultaneously over spoken words and visual context.

By stepping beyond the limits of text-only analysis and embracing multimodal content understanding, the system enhances its sensitivity to high-engagement signals enormously, resulting in a more effective and human-like comprehension of content virality behaviors.

E. Viral Moment Detection With LLMs

After multimodal fusion, every enhanced segment that has both the audio transcription and the visual description combined is fed into a high-level semantic analysis with the support of a large language model (LLM) using LangChain.

The system utilizes the `viral_llm_call` function with a custom system message to ask the LLM to point out segments most likely to be viral. The model is asked to judge the content in terms of a set of engagement-focused metrics such as:

- **Emotional Intensity:** Identification of intense emotional signals like excitement, humor, anger, or empathy, which strongly correlate with watch time and virality.
- **Humor and Irony:** Detection of comedic moments, witty dialogues, or ironic events that might appeal to viewers.
- **Surprise or Contention:** Detection of scenes that contain plot turns, contentious remarks, or visually sensational transitions tending to cause viewer reactions.
- **Relatability and Shareability:** Analysis of content conforming to typical social encounters, cultural allusions, or universally recognized storylines enhancing share likelihood.

Diverging from usual keyword- or frequency-based methodologies, this technique exploits deep semantic thinking to assess the implicit story, emotional arcs, and text-visual correspondence within each snippet. The algorithm is able to process context, sentiment progression, and cross-modal signals and, as a consequence, detect nuance that is normally skipped over by conventional algorithms.

For each identified viral moment, the system also creates a corresponding rationale for why the segment will most likely be able to generate audience engagement. This explanatory level enhances transparency, boosts user confidence in automated picks, and offers valuable analytical feedback that can guide future content creation plans.

In addition, the function outputs a structured format that includes the start and end timestamps, word-level time markers for each viral snippet, the full transcription, and the title of the source video. This metadata is further utilized in the downstream pipeline to automate precise video cutting and editing tasks.

By doing so, the system moves beyond superficial pattern matching and reaches a more human-like, more nuanced assessment of content virality potential.

F. Mapping Viral Moments to Video Timeframes

Once these high-potential viral segments have been singled out through multimodal semantic analysis, the system performs temporal alignment to position the extracted clips exactly in their context within the initial video timeline. This pivotal operation ensures the clips retain the narrative integrity detected by them without being cut short or starting mid-stride.

The mapping task as shown in the figure 2 is coordinated by the `extract_timeframes_for_viral_moments` function, which exploits a mix of:

- **Lexical Matching:** Matching key phrases from the audio transcript of the segment (LLM output) with global word-level transcription.
- **Semantic Anchors:** Leaning on nearby context and visual information to build robustness to small speech variances or scene changes.
- **Timestamp Information:** Applying word-level timing metadata to precisely identify the beginning and end of each identified viral moment.

Through combining both lexical and semantic cues, the system reduces boundary drift the process through which extracted fragments start or stop beyond the original content window thereby maintaining narrative coherence and enhancing the aesthetic value of resulting final video outputs.

In addition, the mapping process is engineered to preserve natural transitions, without mid-sentence cut-offs or visual breaks that might break viewer continuity. This precision alignment is critical to the goal of having the extracted viral clips be contextually self-contained, emotionally engaging, and immediately effective for social media viewers.

G. Subtitle Generation and Formatting

To further improve accessibility and viewer engagement, the system supports dynamic, word-level subtitles projected over the video content. This aspect is most important for mobile-first viewers, where muted video view is common, and visual text prompts play a major role in determining viewer retention and understanding.

Subtitle generation is done with the help of the `moviepy` library, a Python high-level video editing library. The process includes two significant steps:

- **Line Segmentation:** The `split_text_into_lines` function splits the transcript into logically consistent and visually acceptable lines. Segmentation parameters include limits on maximum character count, maximum display time, and natural speech breaks (gaps), to maximize readability and alignment with spoken audio.
- **Animated Caption Overlay:** The `create_caption` function automatically creates time-synced text overlays, where words are sequentially highlighted as they are uttered. This provides a visually rhythmic experience

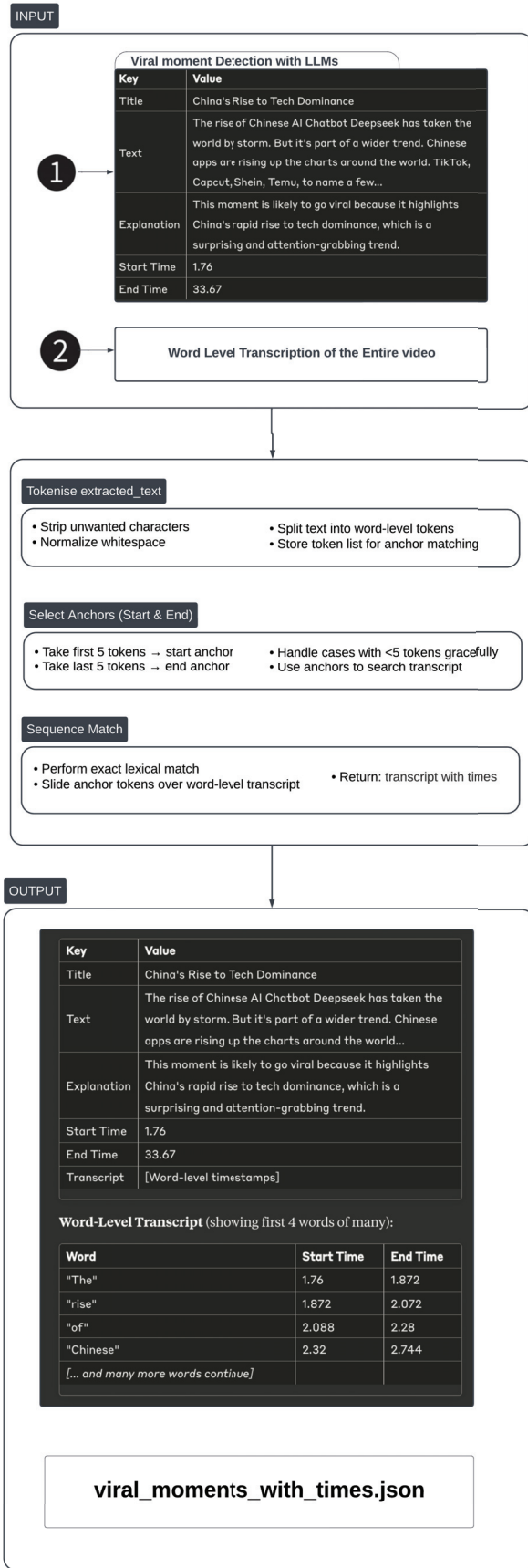


Fig. 2. Mapping Viral Moments to Video Timeframes

synchronized with speech cadence, facilitating comprehension even without audio playback.

Subtitle design favours legible typography, least distraction, and response in real time, conforming to the vertical video format more commonly employed within social media. Highlighting of words at the word level not only enhances readability of the text but also strengthens cognitive engagement through enabling viewers to recognize unconsciously speech flow.

This solution guarantees that the system supports varying user environments such as sound-off mobile scenarios while also increasing interaction rates, watch time, and user satisfaction.

H. Platform Optimal Video Processing

For the purpose of ensuring utmost compatibility and aesthetic appeal on most popular short-form content sites like YouTube Shorts, Instagram Reels, and TikTok, the system concludes all output clips in a 9:16 vertical aspect ratio. This conversion is paramount to mobile-first consumption, where the leading format is vertical video.

The pipeline for video processing makes use of both moviepy and OpenCV to execute a sequence of operations that transform landscape or mixed-aspect-ratio source material into a good-quality vertical output. The central transformation is executed by the process_video_with_subtitles function, which carries out the following steps:

- **Aspect Ratio Conversion:** The frame is converted to 9:16, either through intelligent cropping (region-of-interest centered about faces or motion) or strategic padding to prevent loss of content.
- **Contrast and Brightness Normalization:** Histogram-based methods and gamma correction are used to provide readability under diverse lighting conditions.
- **Frame Stabilization:** Small camera shakes are smoothed through optical flow algorithms in OpenCV, improving visual coherence.
- **Subtitle Integration:** Post-scaling, dynamic word-level subtitles are overlaid to provide accurate alignment with the new frame size and still remain legible.

This pipeline of video transformation ensures that the end product fulfills the social platforms' technical and aesthetic specifications, improves user engagement, and offers uniform view quality across different screen sizes and devices.

I. Metadata logging, and Clip Archival

After the viral clips have been processed and extracted, each is saved with clean, human-readable filenames that mirror their thematic content usually including salient elements like topic keywords, timestamps, or sentiment tags. This systematic naming scheme makes intuitive manual browsing and automated categorization easy.

In parallel, a full metadata logging procedure is performed. For each clip, a systematic JSON or tabular metadata record is produced, capturing vital attributes such as:

- **Temporal Boundaries:** Begin and end times for every viral moment in the source video.
- **Topic Summaries:** Brief semantic summaries from the transcript and visual information that capture the central theme of the clip.
- **Sentiment Scores:** Numerical emotional polarity (e.g., positive, neutral, negative) computed from both text and visual input.
- **Visual Tags:** Designations for primary objects, scenes, or expressions found during the video analysis stage.

This organized metadata has various downstream applications, such as effective content retrieval, semantic indexing, and integration with future ranking functions or recommendations. In addition, the high-level fusion of audio-visual understanding ensures that the metadata remains contextually sensitive, supporting deeper personalization and search relevance in broader media processing pipelines.

J. Web-Based User Interface

For non-technical users, the system provides a simple yet capable web interface as shown in Figure 3, developed based on the Flask framework.

The user-friendly application makes the entire process streamlined, enabling users to:

- Enter YouTube video URLs for processing.
- View the processing pipeline in real time.
- Preview extracted viral clips prior to export.
- Download output clips with subtitles and metadata.

The backend processing is contained in a module, which provides major functionalities through a RESTful API. This abstraction layer separates frontend interactions from the processing engine, allowing for modularity and scalability. Users do not have to deal with intricate scripts or terminal commands, so the tool becomes usable for educators, marketers, influencers, and researchers.

In addition to this, the interface is constructed with responsiveness as a consideration for usability on desktop and mobile environments. Upcoming updates can entail user authentication, analytics dashboards, and integrating cloud storage support for long-term content management.

K. System Novelty and Contributions

This work introduces a complete, end-to-end system for automated viral content creation from long videos. The major innovations and contributions of the system are:

- **Multimodal Viral Moment Detection:** The system combines both audio transcriptions and visual descriptions to detect high-engagement moments. This two-modality approach guarantees that both verbal storytelling and non-verbal visual signals like facial expressions, abrupt scene cuts, or bodily humor are detected and processed.
- **Context-Rich Segment-Level Reasoning:** Rather than shallow heuristics or keyword spotting, the system applies deep contextual reasoning to every short-form segment. This encompasses emotional interpretation, se-

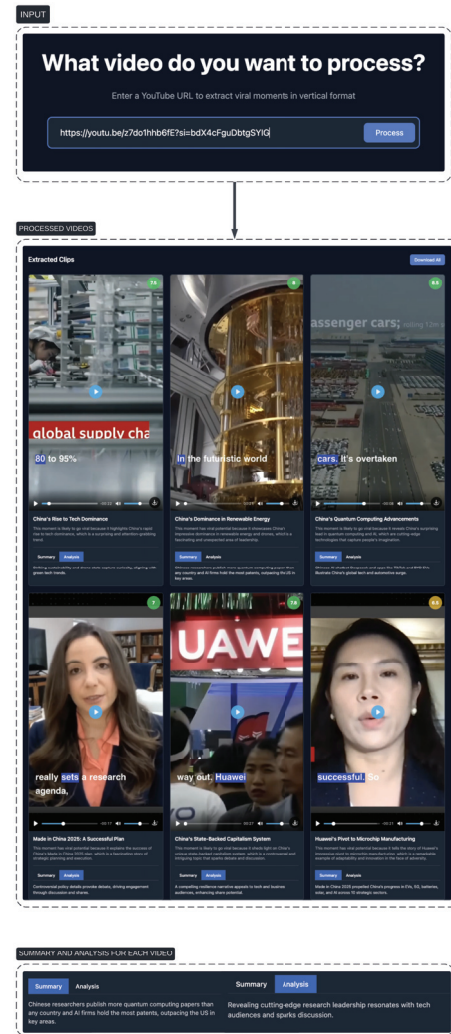


Fig. 3. Web Interface

semantic comprehension, and visual-textual alignment to establish virality potential with greater accuracy.

- **Fully Automated Pipeline:** The architecture does everything from video ingestion and audio transcription to multimodal analysis, segmentation, subtitle rendering, and short-form export automatically. This reduces manual intervention, democratizing the creation of viral content to non-technical or editing-user levels.
- **Platform-Optimized Output:** All outputs are optimized for top vertical-video platforms like YouTube Shorts, TikTok, and Instagram Reels, with the capability to export directly to them without manual formatting, effectively reducing cross-platform content distribution complexity.
- **Scalable and Modular Design:** The system's modular design, such as using REST APIs, asynchronous processing, and plug-and-play model components, makes the system scalable for future integration (e.g., multilingual, user personalization, or recommendation engines).

This research greatly benefits the new area of AI-based

content generation, presenting a new model for scalable, intelligent, and human-centered video repurposing. Through integrating speech, vision, and semantic understanding, the system facilitates efficient and effective conversion of raw videos into well-refined, viral-ready content.

IV. RESULTS AND ANALYSIS

Assessing the effectiveness of an AI system for creating short-form video content is particularly challenging because there are no established standards for measuring creativity, audience engagement, and viral likelihood. A specialized evaluation system was created based on Google's Gemini multimodal LLM as shown in the figure 4. This solution replaces conventional metrics with a qualitative pipeline analysis that emulates expert feedback on video material. In particular, each produced video is passed to the Gemini model via a API, which yields a JSON-formatted response that includes a brief content summary, a qualitative evaluation of the video's potential virality based on engagement metrics, trend relevance, and shareability, and a 0-to-10 "Virality Score" which is also displayed in the web interface.

This process of evaluation showed promising results. A testing sample of AI-created videos yielded virality scores between 6.2 and 9.1, which implies that the system tends to create content with moderate to high online engagement potential consistently. The qualitative results from Gemini showed that videos that used humor, trending audio snippets, visually appealing transitions, or emotionally moving narratives tended to be scored higher. In addition, the summaries generated by the model confirmed that the videos were thematically coherent and narratively structured two aspects most directly related to viewer engagement and platform promotion algorithms. The results demonstrate that the system is able to effectively identify factors behind virality on platforms like TikTok, Instagram Reels, and YouTube Shorts.

However, the methodology of evaluation contains built-in biases. The virality scores and insights generated by Gemini are model-generated and have not been empirically tested against actual audience actions or platform data. The procedure, although innovative, is still partly subjective and could be an echo of inherent biases in the model's training data. Besides, external parameters like timing of upload, demographic of users, and platform-dependent algorithms were ignored, which may greatly influence the actual performance of a video on the internet.

In subsequent versions of this research, the evaluation framework can be improved by adding human feedback in the form of surveys or expert scores, A/B testing across social media platforms, and crowd-sourced assessment integration to expand the objectivity of the evaluation. In spite of these constraints, the existing evaluation strategy gives a significant starting point for exploring the creative and viral potential of AI-generated short-form video content.

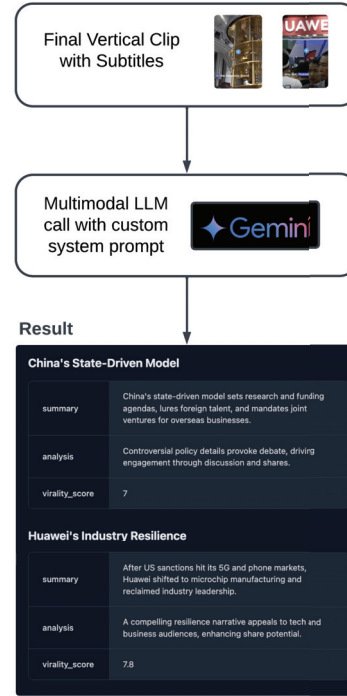


Fig. 4. Gemini evaluation

V. CONCLUSION

This work introduced an AI-based system for automated short-form social media video generation and evaluation with the goal of simplifying content creation and maximizing virality. Through the use of generative AI methods, the system can generate coherent, interesting, and aesthetically pleasing video content with little human intervention. The incorporation of a Gemini-based evaluation pipeline also allowed for a new, qualitative measurement of each video's viral appeal, providing formatted feedback on engagement metrics, trend relevance, and shareability.

Experimental outcomes, while assessed via a non-standardized model-aided approach, showed that most of the generated videos had moderate to high virality potential. This justifies the potential of AI-based content generation pipelines in fast-changing digital environments. While the lack of human-based rating and live platform testing continues to be a shortcoming, the presented system lays a solid basis for further research in creative automatization.

Finally, this study advances the emergent field of artificial intelligence for digital media content production. It introduces avenues toward large-scale, smart content generation software that will support influencers, advertisers, and media outlets to produce timely and effective content affordably. Possible extensions of the study in the future could integrate consumer feedback, deployment across platforms, and trend-based real-time adjustments to extend the system's pertinence and utility to actual situations.

REFERENCES

- [1] Zhou, Peng Yuan & Wang, Lin & Liu, Zhi & Hao, Yanbin & Hui, Pan & Tarkoma, Sasu & Kangasharju, Jussi. (2024). A Survey on Generative AI and LLM for Video Generation, Understanding, and Streaming. doi: 10.13140/RG.2.2.20978.89283.
- [2] Bryan Wang & Yuliang Li & Zhaoyang Lv & Haijun Xia & Yan Xu & Raj Sodhi. LAVE: LLM-Powered Agent Assistance and Language Augmentation for Video Editing, 2024, <https://arxiv.org/abs/2402.10294>.
- [3] Cao, M., Zheng, J. & Zhang, C. AI-based Chinese-style music generation from video content: a study on cross-modal analysis and generation methods. J AUDIO SPEECH MUSIC PROC. 2025, 8 (2025). doi: 10.1186/s13636-025-00397-3.
- [4] Sadique, P.M.A., Aswiga, R.V. Automatic summarization of cooking videos using transfer learning and transformer-based models. Discov Artif Intell 5, 7 (2025). doi: 10.1007/s44163-025-00230-y.
- [5] Zhicheng Geng & Luming Liang & Tianyu Ding & Ilya Zharkov. RSTT: Real-time Spatial Temporal Transformer for Space-Time Video Super-Resolution. 2022. <https://arxiv.org/abs/2203.14186>.
- [6] Chen, Guo & Zheng, Yin-Dong & Jiahao, Wang & Xu, Jilan & Huang, Yifei & Pan, Junting & Wang, Yi & Wang, Yali & Qiao, Yu & Lu, Tong & Wang, Limin. VideoLLM: Modeling Video Sequence with Large Language Models. (2023). doi: 10.48550/arXiv.2305.13292.
- [7] Pang, Zongshang & Otani, Mayu & Nakashima, Yuta. Measure Twice, Cut Once: Grasping Video Structures and Event Semantics with LLMs for Video Temporal Localization. (2025). doi: 10.48550/arXiv.2503.09027.
- [8] Maaz, Muhammad & Rasheed, Hanoona & Khan, Salman & Khan, Fahad. Video-ChatGPT: Towards Detailed Video Understanding via Large Vision and Language Models. (2024). 12585-12602. doi: 10.18653/v1/2024.acl-long.679.
- [9] Nan Zhuang, Zhenyuan Ma, Yujie Zhou, Xinyi Li, Pinhao Wang, Zhiyuan Huang, Shuo Zhai, and Fangtian Ying. "Alleviating Elderly's Medical Communication Issue with Personalized LLM-Generated Short-Form Video." Open Access by IOS Press, 2024, pp. 763-772. doi: 10.3233/FAIA240078.
- [10] Tiezheng Yu, Hanchao Yu, Davis Liang, Yuning Mao, Shaoliang Nie, Po-Yao Huang, Madian Khabsa, Pascale Fung, and Yi-Chia Wang. 2023. Generating Hashtags for Short-form Videos with Guided Signals. In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 9482–9495, Toronto, Canada. Association for Computational Linguistics.
- [11] A. Singh, "A Survey of AI Text-to-Image and AI Text-to-Video Generators," 2023 4th International Conference on Artificial Intelligence, Robotics and Control (AIRC), Cairo, Egypt, 2023, pp. 32-36, doi: 10.1109/AIRC57904.2023.10303174.
- [12] Ipsita Pattnaik, Pulkit Narwal. Implicit embedding based multi modal attention network for Cricket video summarization. Engineering Applications of Artificial Intelligence. Volume 148. 2025. 110428, ISSN 0952-1976, doi: 10.1016/j.engappai.2025.110428.
- [13] Angela John and Theophilus Aidoo and Hamayoon Behmanush and Irem B. Gunduz and Hewan Shrestha and Maxx Richard Rahman and Wolfgang Maaß, LLMRS: Unlocking Potentials of LLM-Based Recommender Systems for Software Purchase, john2024llmrsunlockingpotentialsllmbased, 2024, <https://arxiv.org/abs/2401.06676>.
- [14] K. Rho, H. Lee, V. Iverson and J. S. Chung, "LAVCap: LLM-based Audio-Visual Captioning using Optimal Transport," ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Hyderabad, India, 2025, pp. 1-5, doi: 10.1109/ICASSP49660.2025.10888241.
- [15] Ilaslan, M. F., Köksal, A., Lin, K. Q., Satar, B., Shou, M. Z., & Xu, Q. (2025). VG-TVP: Multimodal Procedural Planning via Visually Grounded Text-Video Prompting. Proceedings of the AAAI Conference on Artificial Intelligence, 39(4), 3886-3894. <https://doi.org/10.1609/aaai.v39i4.32406>
- [16] Aditi Singh, "A Survey of AI Text-to-Image and AI Text-to-Video Generators", 4th International Conference on Artificial Intelligence, Robotics and Control (AIRC), IEEE, 2023, <http://dx.doi.org/10.1109/AIRC57904.2023.10303174>.
- [17] Song S, Yang C, Xu L, Shang H, Li Z, Chang Y. TravelRAG: A Tourist Attraction Retrieval Framework Based on Multi-Layer Knowledge Graph. ISPRS International Journal of Geo-Information. 2024; 13(11):414. <https://doi.org/10.3390/ijgi13110414>
- [18] K. Sreekar, V. Aswin, V. Narayanan, S. K. Thangavel, K. Somasundaram and S. K. Shanmugam, "MedBot – An NLP based ChatBot for Diabetes Prediction," 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kamand, India, 2024, pp. 1-11, doi: 10.1109/ICCCNT61001.2024.10724179.